# Experiment No.: 6

# Title:

**Design Responsive Login Form using HTML5, CSS3 and Bootstrap with JavaScript Validation using DOM Manipulation**

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## 1. Aim

To design and develop a responsive login form using **HTML5, CSS3, and Bootstrap**, and implement **client-side validation** using **JavaScript and DOM manipulation**.

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## 2. Learning Objectives (LO Mapping)

**LO1:** Understand web development fundamentals.
**LO3:** Develop frontend interfaces using HTML, CSS, and JavaScript.
**LO6:** Use modern tools and best practices for UI development.

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## 3. Software Requirements

Web Browser (Google Chrome / Mozilla Firefox / Edge)
Visual Studio Code (VS Code)
Bootstrap CDN
Operating System: Windows / Linux / macOS

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## 4. Introduction

A login form is one of the most essential components of modern web applications. It allows users to authenticate themselves securely before accessing system resources.
Responsive design ensures that the interface adapts automatically to different screen sizes such as mobile phones, tablets, and desktops.
Technologies used:

- **HTML5** → Structure of webpage
- **CSS3** → Styling and layout
- **Bootstrap** → Responsive framework
- **JavaScript** → Validation and interactivity
- **DOM Manipulation** → Dynamic control of page elements

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## 5. Theory

### 5.1 HTML5 Forms

HTML provides form elements such as:

- `<form>`
- `<input>`
- `<label>`
- `<button>`

Common input types:

- text
- email
- password

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### 5.2 CSS3

CSS is used for:

- Layout styling
- Alignment
- Colors and fonts
- Responsive appearance

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### 5.3 Bootstrap

Bootstrap is a frontend framework that provides:

- Grid system
- Responsive layout
- Predefined UI components
- Mobile-first design

Example classes:

- `container`
- `row`
- `col-md-6`
- `form-control`
- `btn`

## What is Bootstrap?

**Bootstrap** is a **free and open-source front-end framework** used to design **responsive and mobile-friendly websites** quickly using **HTML, CSS, and JavaScript**.
It provides **ready-made components**, styles, and layout systems so developers do not need to write CSS from scratch.

## Definition

**Bootstrap is a CSS framework that helps developers create responsive and visually attractive web pages using predefined classes and components.**

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## Why Bootstrap is Used?

### 1. Responsive Web Design

Bootstrap automatically adjusts website layout for:

- Mobile 📱
- Tablet 💻
- Desktop 🖥️

Example:

```
<div class="container"></div>
```

The layout adapts to screen size automatically.

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### 2. Faster Development

No need to design everything manually.
Bootstrap provides:

- Buttons
- Forms

- Navigation bars
- Cards
- Tables
- Alerts

Example:

```
<button class="btn btn-primary">Login</button>
```

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## 3. Grid System (Layout Management)

Bootstrap uses a **12-column grid system** to create structured layouts.

Example:

```
<div class="row">
  <div class="col-md-6">Left</div>
  <div class="col-md-6">Right</div>
</div>
```

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## 4. Predefined Styling

Provides built-in styles for:

- Colors
- Spacing
- Typography
- Alignment

Example:

text-center
bg-primary
mt-3

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## 5. Cross-Browser Compatibility

Works properly on:

- Chrome
- Firefox
- Edge
- Safari

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## 6. Easy Form Design

Bootstrap makes professional forms easily.

Example:

```
<input type="email" class="form-control">
```

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## 7. Built-in JavaScript Components

Interactive elements like:

- Modals
- Dropdowns
- Carousel
- Navbar toggle

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## How Bootstrap Works

HTML Structure
↓
Bootstrap CSS Classes
↓
Automatic Styling + Responsiveness
↓
Professional UI Design

## Simple Example

```html
<!DOCTYPE html>
<html>
<head>
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
</head>

<body class="container mt-5">

<h2 class="text-center">Login</h2>
<button class="btn btn-success">Submit</button>

</body>
</html>
```

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### 5.4 DOM Manipulation

## What is DOM? (Document Object Model)

The **DOM (Document Object Model)** is a **programming interface** that represents an HTML or XML document as a **tree structure of objects**, allowing JavaScript to **access, modify, add, or delete** elements dynamically.

## How DOM Work

When a web page loads:

HTML Document
↓
Browser converts HTML into
↓
DOM Tree (Objects)
↓
JavaScript can access & modify elements

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## DOM Tree Example

HTML Code

```html
<html>
  <body>
    <h1>Hello</h1>
    <p>Welcome</p>
```

```
  </body>
</html>
```

# DOM Structure

```
Document
 └── html
      └── body
           ├── h1
           └── p
```

Each element becomes an **object (node)**.

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## Why DOM is Used

DOM allows developers to:

✅ Change webpage content
✅ Change styles dynamically
✅ Handle user events
✅ Validate forms
✅ Add or remove elements

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## Common DOM Operations

### 1. Select Elements

```
document.getElementById("demo");
```

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### 2. Change Content

```
document.getElementById("demo").innerHTML = "Hello World";
```

---

### 3. Change Style

```
document.getElementById("demo").style.color = "red";
```

---

### 4. Handle Events

```
button.onclick = function() {
  alert("Clicked!");
};
```

---

## DOM in Login Form

When user clicks **Login**:

```
User Input
   ↓
JavaScript reads values using DOM
   ↓
Validation performed
   ↓
Error/Success message shown
```

Example:

```
let email = document.getElementById("email").value;
```

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### Key DOM Terms

| Term | Meaning |
|---|---|
| Document | Entire webpage |
| Element | HTML tag |
| Node | Object in DOM tree |
| Event | User action (click, input) |
| Manipulation | Changing webpage dynamically |

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### Advantages of DOM

✓ Makes web pages interactive
✓ Dynamic content updates
✓ Enables form validation
✓ Improves user experience

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### 5.5 JavaScript Validation

Client-side validation prevents incorrect data submission.
Examples:

- Empty field checking
- Email format validation
- Password length validation

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## 6. Diagram

### Login Form Architecture

User Input
↓
HTML Form
↓
JavaScript Validation
↓
DOM Manipulation
↓
Success / Error Message

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## 7. Source Code

### Step 1: Create File

Create a file named: “login.html”

### Step 2: HTML + Bootstrap + JavaScript Code

```
<!DOCTYPE html>
<html>
<head>
```

```html
    <title>Responsive Login Form</title>
    <!-- Bootstrap CDN -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
    <style>
        body {
            background-color: #f2f2f2;
        }
        .login-box {
            margin-top: 100px;
            padding: 30px;
            background: white;
            border-radius: 10px;
            box-shadow: 0px 0px 10px gray;
        }
    </style>
</head>
<body>
<div class="container">
    <div class="row justify-content-center">
        <div class="col-md-5 login-box">
            <h3 class="text-center">Login Form</h3>
            <form onsubmit="return validateForm()">
                <div class="mb-3">
                    <label>Email</label>
                    <input type="text" id="email" class="form-control">
                    <small id="emailError" class="text-danger"></small>
                </div>
                <div class="mb-3">
                    <label>Password</label>
                    <input type="password" id="password" class="form-control">
                    <small id="passError" class="text-danger"></small>
                </div>
                <button type="submit" class="btn btn-primary w-100">
                    Login
                </button>
                <p id="successMsg" class="text-success text-center mt-3"></p>
            </form>
```

```
        </div>
    </div>
</div>
<script>
function validateForm() {
    let email = document.getElementById("email").value;
    let password = document.getElementById("password").value;
    let valid = true;
    document.getElementById("emailError").innerHTML = "";
    document.getElementById("passError").innerHTML = "";
    document.getElementById("successMsg").innerHTML = "";
    if(email === "") {
        document.getElementById("emailError").innerHTML =
            "Email is required";
        valid = false;
    }
    if(password.length < 6) {
        document.getElementById("passError").innerHTML =
            "Password must be at least 6 characters";
        valid = false;
    }
    if(valid) {
        document.getElementById("successMsg").innerHTML =
            "Hi Sucheta, Your User's Login Successful!";
    }
    return false;
}
</script>
</body>
</html>
```

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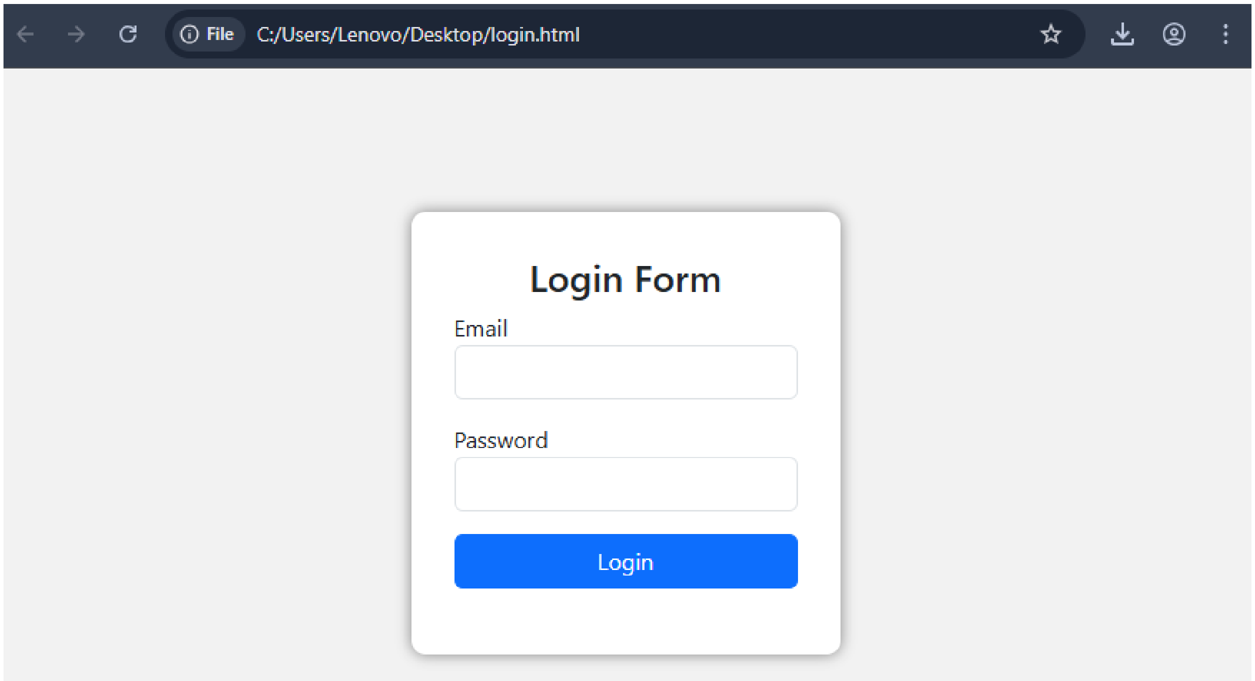
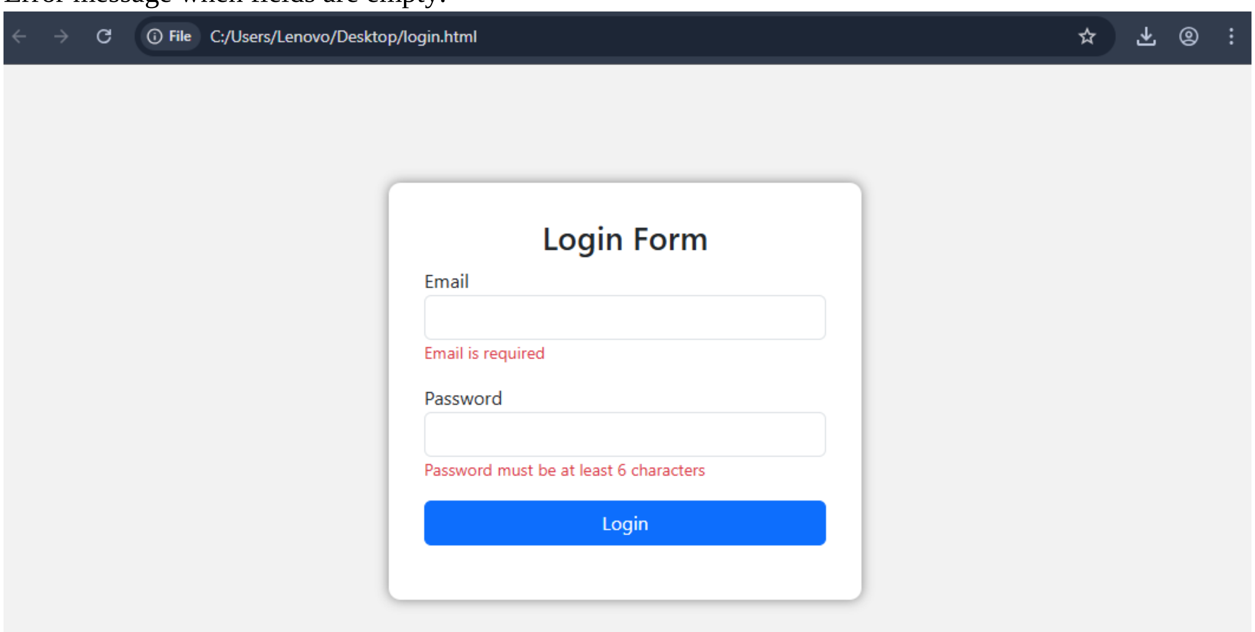
# 8. Output

## Expected Output Screens

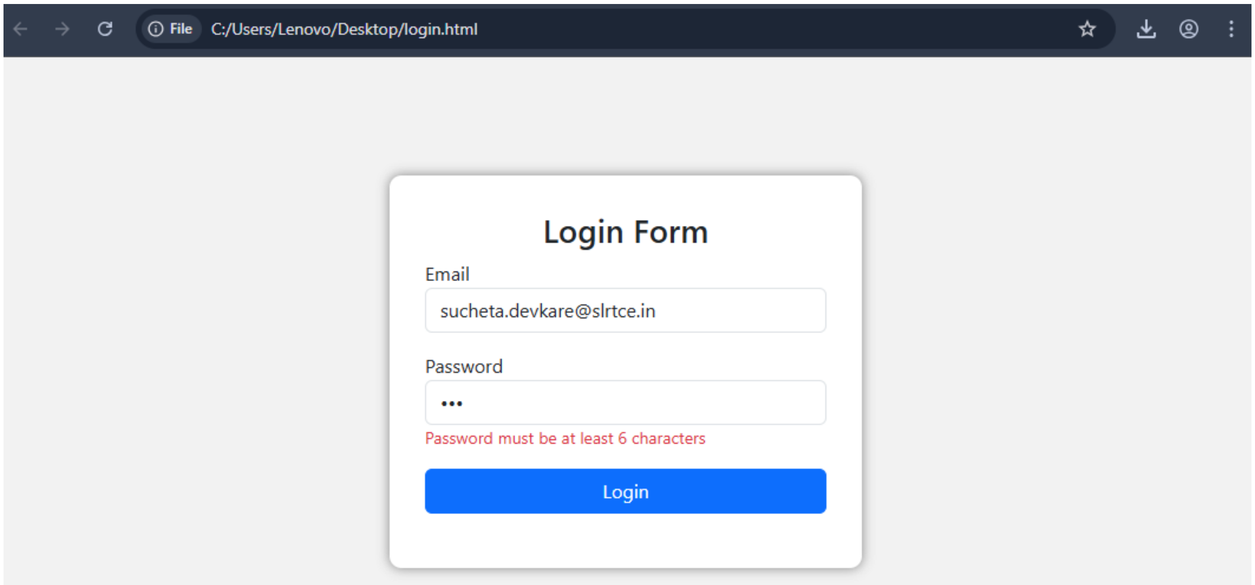
1. Responsive login form displayed at center.



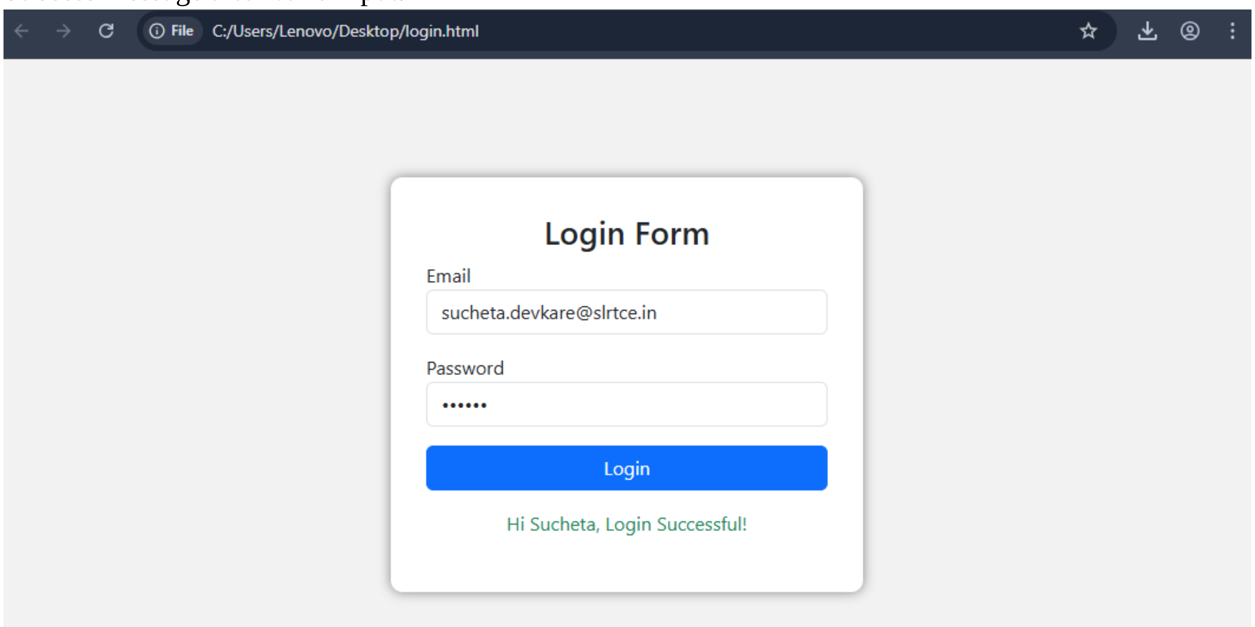

2. Error message when fields are empty.




3. Password validation message.




4. Success message after valid input.

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# 9. Result

The responsive login form was successfully designed using HTML5, CSS3, and Bootstrap, and input validation was implemented using JavaScript and DOM manipulation.

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# 10. Conclusion

This experiment demonstrates how frontend technologies work together to create interactive and responsive web interfaces. JavaScript validation improves user experience by preventing invalid data submission.

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# 11. Viva Questions

1. What is HTML5 form validation?
2. What is Bootstrap?
3. Define responsive web design.
4. What is DOM in JavaScript?
5. Difference between client-side and server-side validation.
6. What is getElementById()?
7. Why is JavaScript validation required?
8. What is a CDN?
9. Explain Bootstrap grid system.
10. What is event handling in JavaScript?